Department of Computer Science and Engineering
University of Barisal
3rd Year 2nd Semester Final Examination, 2024

Course Code: CSE-3203 **Course Title: Theory of Computation**

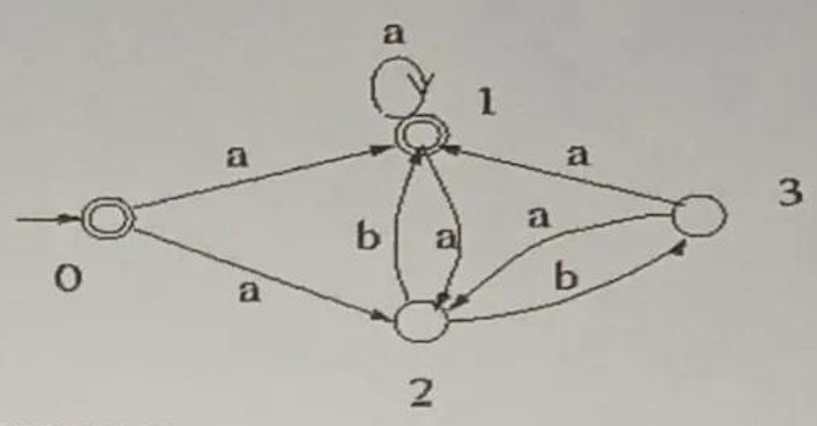
Total time: 3.00 hours **Total marks: 60.0**

(Note: Answer any five set of questions from the followings)

1. a) Define Turing Machine. Explain pushdown automaton. 4
   b) Prove that every language accepted by a multitap TM is recursively enumerable. 6
   c) What do you know about context-free grammars? 2

2. a) Define with examples: Alphabet, String, Language 6
   b) Give a regular expression for the following language B over the alphabet {a, b}. 4
   $B = \{w \mid w \text{ does not contain the substring aaa}\}$.
   c) When is an expression called regular expression? 2

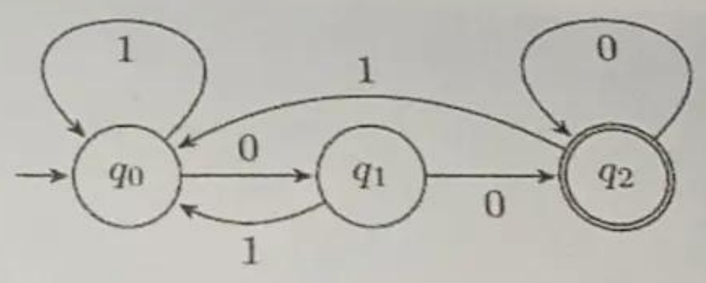
3. a) Use the pumping lemma to prove that the language $A = \{ 0^{2n} 1^{3n} 0^{n} \mid n \geq 0 \}$ is not context free. 3
   b) Convert the following NFA to DFA. 6



   c) State the Pumping Lemma for regular languages. 3

4. a) Summarize the principal closure properties for regular languages. 3
   b) Convert DFA's to regular expressions by eliminating states (using an arbitrary example). 5
   c) Prove that if L is a regular language over alphabet Σ, then $\bar{L} = \Sigma* - L$ is also a regular language. 4

5. a) Construct a context-free grammar for the following DFA: 5



   b) Show that the grammar ({S}, {a, b}, R, S) with rules R = S → aS | aSbS | ϵ is ambiguous. 4
   c) Explain inverse homomorphism. 3

6. a) Elaborate the three operations on languages that the operations of regular expression represent. 4
   b) Prove that if we add a finite set of strings to a regular language, the result is a regular language. 5
   c) Write the closure properties of regular languages. 3

7. a) Prove that the following are not regular languages: 2+2
   (i) The set of strings of 0's and 1's beginning with 1, such that when interpreted as an integer, that integer is a prime.
   (ii) The set of strings of the form $0^i 1^j$ such that the greatest common divisor of i and j is 1.
   b) State the properties of a parse tree. Construct a parse tree for the following string: 4
   $S \rightarrow S\,S+ \mid S\,S* \mid a$
   c) Prove that every language defined by a regular expression is also defined by a finite automata. 4

8. a) State formal definition of a finite automata. Give an example. 4
   b) Consider the regular expression $(a(cd)^*b)^*$. 2+2
   (i) Find a string over $\{a, b, c, d\}^4$ which matches the expression.
   (ii) Find a string over $\{a, b, c, d\}^4$ which does not match the expression.
   c) Build an ε-NFA for the following language: 4

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